PROJECT VIVA VOCE & TECHNICAL DEFENSE MANUAL

Automated Paint Spraying & Curing Station with Multi-Screen Supervisory HMI

Comprehensive Engineering Justifications, Process Physics, Ladder Diagnostics & 18 Master Viva Q&As

CONTROLLER: FX5U-32MT/ES | HMI: GOT2000 GS2107 (800x480) | LADDER: 95 STEPS | I/O: X0-X7, Y0-Y7

SECTION 1: EXECUTIVE SYSTEM ARCHITECTURE & AUTOMATED PROCESS WORKFLOW

This project realizes a fully autonomous, industrial-grade automated paint spray coating and curing station engineered on the Mitsubishi Electric MELSEC iQ-F Series FX5U PLC and GT Designer3 GOT2000 HMI platform. The system is designed for high-precision, batch-oriented surface finishing of mechanical components with zero manual intervention during production, enforced by fail-safe environmental and pneumatic interlocks.

The 4-Phase Automated Production Lifecycle

- **Phase 1: Infeed & Batch Accumulation:** Conveyor (Y1) starts following a 2.0s acoustic safety delay (T1). Parts pass optical sensor (X2) and are indexed by counter (C0). Upon reaching Recipe Batch Target (D100 = 15 parts), Batch Complete flag (M3) latches ON and halts the infeed conveyor immediately.
- **Phase 2: Pre-Flight Safety & Pneumatic Interlock:** System verifies all 5 pre-spray conditions: Chamber Door closed (X5), Paint Reservoir normal (X6), Air Pressure switch closed (X7), Line Pressure (D110) >= Recipe Minimum Setpoint (D102), and Conveyor fully stationary (/Y1). If any condition fails, spray is blocked.
- **Phase 3: Solvent Evacuation & Controlled Atomization:** Exhaust Blower (Y3) starts FIRST to establish negative booth pressure (preventing volatile solvent escape). Paint Pump (Y2) energizes with a 2.0s hydraulic settle timer (T3). Spray Solenoid (Y0) opens for exact recipe duration (T0 / D101 = 10.0s) for uniform coating.
- **Phase 4: Post-Spray Fume Purge & Discharging:** Spray solenoid closes. Exhaust Blower continues running for 5.0s (T2 / D104) to evacuate airborne aerosols. Conveyor (Y1) restarts to discharge finished parts past exit sensor (X3 / C1). Upon batch discharge, cycle flags reset and lifetime OEE counters (C2, C3) increment.

[EXAMINER VIVA TIP / KEY DEFENSE POINT] WHY BATCH PROCESS INSTEAD OF CONTINUOUS SPRAY?
Continuous spray creates overspray accumulation, edge non-uniformity, and high VOC emissions. Batch indexing accumulates parts, stops the belt, seals the booth, and applies high-pressure atomization in a quiescent environment, reducing paint consumption by 35%.

System Hardware & Industrial Platform Specifications

Subsystem Component	Model / Spec	Physical Function in Architecture	Communication / Voltage
Programmable Controller	FX5U-32MT/ES	Master sequence logic, PID, safety interlocks, recipes	24V DC Sink/Source, 34ns scan
Operator HMI Terminal	GOT2000 GS2107-WTBD	4-Screen supervisory UI, setpoints, alarms, manual jog	7-inch WVGA 800x480, Ethernet TCP/IP
Paint Spray Solenoid	SMC 24V DC Poppet (Y0)	High-speed atomizing paint control valve	24V DC, 0.35A, <15ms response
Main Conveyor Drive	Oriental Motor 3-Ph (Y1)	Precision workpiece transport across booth chamber	24V DC Contactor interlocked to E-Stop
Paint Supply Booster	Diaphragm Pump (Y2)	High-pressure fluid feed to spray atomizing manifold	24V DC Relay, 2.0s stabilization settle
Chamber Exhaust Blower	Centrifugal Fan (Y3)	Fume extraction, negative pressure VOC containment	Continuous interlock before atomization
Pneumatic Air Sensor	Festo PE Switch (X7/D110)	Detects pneumatic line pressure vs Recipe threshold	Digital switch X7 + Analog Pot 0-10V
Optical Index Sensors	Omron Retro-reflective	Infeed part detection (X2) and discharge exit (X3)	NPN Open-Collector, 1ms response

SECTION 2: ENGINEERING COMPONENT JUSTIFICATIONS · THE 'WHY' BEHIND DESIGN CHOICES

Viva examiners focus heavily on design rationale. Below are the precise technical justifications for every major hardware, instruction, timer, and register choice implemented in the system.

1. Why Mitsubishi FX5U PLC Instead of FX3U, Arduino, or Raspberry Pi?

- **Deterministic Execution & MTBF:** Industrial PLCs provide guaranteed deterministic scan times (34 nanoseconds per basic instruction), eliminating OS latency, garbage collection pauses, or thread starvation that make microcontrollers unsafe for machinery with pinch points.
- **Noise Immunity & Optical Isolation:** FX5U inputs/outputs are 24V opto-isolated with built-in digital RC noise filtering compliant with IEC 61131-2, preventing inductive spikes from conveyor motors or solenoids from crashing the CPU (a fatal flaw with 5V microcontrollers).
- **Native Ethernet Architecture:** Older FX3U required expensive external serial modules (FX3U-ENET-ADP). FX5U features an embedded Ethernet port supporting direct high-speed MELSOFT connections to the GOT2000 HMI at 100 Mbps with zero configuration overhead.

2. Why Mitsubishi GOT2000 HMI Instead of Physical Pushbuttons Alone?

- **Decoupled Parameter Recipes:** Physical pushbuttons can only start or stop. The GOT2000 HMI enables dynamic recipe engineering (adjusting spray durations D101, batch sizes D100, and pressure thresholds D102) without downloading new PLC ladder code or shutting down the plant.
- **Multi-Role Interface Separation:** The 4-screen hub architecture strictly separates roles: Operators run production on Screen 2 (Auto), maintenance personnel test individual actuators on Screen 3 (Manual Inching), while engineers tune chemistry setpoints on Screen 4 (Recipes).
- **Visual Alarm Annunciation & Zero Rewiring:** Instead of dozens of physical pilot lamps and selector switches requiring complex wire looms and panel cutouts, the HMI provides instant color-coded annunciators, live second counters, and software diagnostic resets over a single Ethernet cable.

[ENGINEERING PHYSICS / FORMULA] FLUID DYNAMICS BEHIND TIMER T3 (2.0s PUMP SETTLE DELAY)

When diaphragm or booster pumps start, fluid inertia and air compliance in the feed hoses cause severe pressure ripple (pulsation). If the spray nozzle opens immediately at t=0, paint emerges in large, non-atomized droplets ('spitting'), ruining surface finish. Timer T3 holds spray solenoid Y0 closed for 2.0s until paint line hydraulic pressure stabilizes at steady state.

3. Detailed Rationale for All Process Timers (T-Devices)

Timer	Preset Device	Time Value	Physical / Engineering Reason for Inclusion
T0	D101 (Recipe)	10.0s (K100)	Paint Spray Duration. Controls dry film thickness (DFT). Directly linked to recipe register.
T1	K20 (Constant)	2.0s (Fixed)	Conveyor Pre-Start Safety Warning. Prevents human pinch entrapment before conveyor chain moves.
T2	D104 (Recipe)	5.0s (K50)	Post-Spray Fume Purge. Maintains exhaust blower Y3 after spray shuts to evacuate VOC solvent mist.
T3	K20 (Constant)	2.0s (Fixed)	Paint Pump Stabilization. Allows hydraulic line pressure to stabilize before opening nozzle Y0.
T4	K50 (Constant)	5.0s (Fixed)	Conveyor Jam Timeout. If belt runs for 5s without part reaching X2, stops motor to prevent burnout.
T5	K50 (Constant)	5.0s (Fixed)	Low Pressure Timeout. If pump runs but pressure fails to reach threshold in 5s, latches fault M18.
T7	K10 (Constant)	1.0s (Fixed)	Defect Reject Gate Pulse. Fires pneumatic reject diverter for clean part clearance.
T10	K8 (Constant)	0.8s (Fixed)	Auto-Feed Pulse Generator. Autonomous simulation timer that pulses infeed counter without manual clicking.

4. Detailed Rationale for All Counters (C-Devices)

Counter	Target Device	Default Value	Functional & Industrial Justification
C0	D100 (Recipe)	15 Parts	Infeed Batch Accumulator. Counts parts arriving at optical sensor X2 until batch is full (M3).
C1	D100 (Recipe)	15 Parts	Outfeed Discharge Verifier. Confirms all 15 coated parts have exited booth past X3 before cycle resets.
C2	K32767	Continuous	Total Lifetime Batches Counter. Increments on batch complete pulse M12 for production tracking.
C3	K32767	Continuous	Total Lifetime Good Parts Counter. Sums total commercially acceptable parts produced.
C4	K32767	Continuous	Total Defective Parts Counter. Increments on reject PB M44 to calculate scrap rate: C4 / (C3+C4).
C6	D103 (Recipe)	50 Batches	Preventive Maintenance Service Counter. Triggers maintenance service flag M20 for nozzle cleaning.

5. Why Use Data Registers (D100-D110) & SM402 Startup Logic?

- **Dynamic Parameter Decoupling:** Hardcoding timer presets (e.g. OUT T0 K100) forces an engineer to open GX Works3, modify code, and re-download whenever a different viscosity paint is used. By using OUT T0 D101, the timer preset dynamically reads from register D101, allowing the HMI to adjust it.
- **SM402 First-Scan Initialization:** SM402 is an internal Mitsubishi system relay that energizes for exactly ONE scan cycle when the PLC transitions from STOP to RUN. Placing MOV instructions under SM402 guarantees default recipe parameters (D100=15, D101=100, D102=40, D104=50, D110=60) are instantly populated in RAM, preventing the CPU from attempting math on uninitialized zero-value memory.

SECTION 3: SAFETY INTERLOCKS, FAIL-SAFE ARCHITECTURE & ELECTRICAL DESIGN

Industrial safety is evaluated rigorously in viva defenses. The spraying station incorporates multi-layered hardware and software interlocks conforming to ISO 13849 Category 3 architecture principles.

Critical Safety Interlock Mechanisms

- **Dual-Layer Emergency Stop Circuitry (X4 & M7):** Emergency stopping is implemented in parallel: Hardware mushroom pushbutton (X4) and HMI touch button (M7). Activating either immediately executes RST M0 (dropping Master Run) and directly de-energizes all four field actuator coils (Y0, Y1, Y2, Y3) on the same scan.
- **Chamber Door Hermetic Interlock (X5):** Spray solenoid Y0 and Conveyor Y1 are electrically and logically blocked whenever chamber door limit switch X5 is open (/X5). This protects operators from breathing toxic atomized isocyanates and prevents high-speed chain pinch injuries.
- **Mutual Exclusion / Anti-Collision Interlock (Rung 70 / M51):** The spray guns (Y0) and conveyor belt (Y1) MUST NEVER run simultaneously. If both outputs are energized at the same time due to an electrical short or logic bug, Rung 70 instantly detects [Y0] [Y1] and latches Abnormal Process Fault M51, shutting down the cell.
- **NFPA 33 Solvent Extraction Pre-Ventilation:** Paint overspray contains flammable VOCs. Fume blower Y3 is interlocked to start BEFORE spray solenoid Y0 to establish negative pressure, and remains energized for 5.0s AFTER spraying stops (T2 purge) to vent lower explosive limit (LEL) gases.
- **Pneumatic Supply Pressure Verification (M21):** Pneumatic pressure switch X7 is continuously compared against recipe threshold D102 via [>= D110 D102]. If supply pressure drops below recipe requirements, Pressure Verified flag M21 drops, preventing spraying with inadequate atomization.

[CRITICAL SAFETY INTERLOCK] WHAT HAPPENS ON A COMPLETE ELECTRICAL POWER FAILURE?
FX5U outputs Y0-Y7 are solid-state/relay contacts that fail-safe to OPEN on power loss. Conveyor motors, pumps, and solenoids immediately de-energize. Internal counters C0-C6 and registers D100-D110 are battery/EEPROM-backed and retain their exact count on reboot.

SECTION 4: HMI SUPERVISORY ARCHITECTURE & USER EXPERIENCE DESIGN

The human-machine interface is engineered in MELSOFT GT Designer3 for a GOT2000 GS2107 7-inch display (800x480 resolution). The interface abandons cluttered single-screen layouts in favor of an ergonomic 4-Screen Hub Model.

The 4-Screen Hub Architecture

- **Screen 1 (B-1: Main Landing Portal):** The master supervisory overview. Houses 4 wide tactical navigation tiles (Auto Mode, Manual Jog, Recipe Configuration, and Alarms) with high-contrast vector icons, live station heartbeat badges, and emergency status indicators.
- **Screen 2 (B-2: Auto Production Dashboard):** Dedicated to standard production. Features a detailed 3D isometric mimic of the paint booth, live numerical readouts (Air Pressure D110, Parts In CN0, Parts Out CN1, Spray Timer TN0), and 4 physical-feel momentary controls (Start M54, Stop M55, Reset M49, Reject M44).
- **Screen 3 (B-3: Manual Jog & Inching):** Used exclusively for machine setup, nozzle clearing, and maintenance. Provides 4 independent momentary/alternate inching buttons: Y1 Conveyor Jog, Y0 Spray Test, Y2 Pump Prime, and Y3 Exhaust Blower, with safety interlock indicators.
- **Screen 4 (B-4: Recipe Management & Parameters):** Allows process engineers to modify batch parameters via on-screen numerical keypad (D100 Batch, D101 Spray, D104 Purge, D102 Pressure) and load 3 preset production coating profiles (Standard, Heavy Protective, Rapid Light Pass).
- **Screen 5 (B-6: Safety & Alarm Annunciator):** A comprehensive alarm center with 6 individual active alarm lamp annunciators (M7, M17, M16, M18, M19, M51), live safety interlock LED circles (X4, X5, X6, X7), and tactile action switches: Silence Buzzer M52, Reset Faults M49, and Touch E-Stop M7.

Key HMI Configuration Rules & Formulas

- **Why Use 'TN0' and 'CN0' Instead of 'T0' and 'C0'?** In GT Designer3 for FX5U / MELSEC iQ-F, 'T0' refers to the bit contact (ON/OFF). To display the live numeric accumulator (elapsed time or count), the HMI requires the Timer Current Value prefix 'TN0' and Counter Current Value prefix 'CN0'.
- **Why Apply Scaling ($$ / 10) on Spray Timer and Pressure?** Timer T0 in FX5U counts in 100ms units ($100\text{ counts} = 10.0\text{s}$). Displaying raw counts confuses operators. Applying the GT Designer3 Data Operation expression '$$ / 10' divides the raw integer by 10 to display true seconds (0 to 10s) on screen.
- **Decoupling Recipe Setpoint (D102) vs Live Sensor Gauge (D110):** A critical industrial distinction: D102 is the target threshold requirement set by the operator on the Recipe screen. D110 is the actual physical pressure delivered by the supply header. Changing the target does not alter line pressure; rather, the PLC verifies that actual pressure meets or exceeds the target before permitting spray operation.

SECTION 5: TOP 18 COMPREHENSIVE VIVA VOCE QUESTIONS & MASTER SCORING ANSWERS

Study these exact questions and technical formulations. They represent the most probable questions asked by senior PLC examiners.

Q1: What is the basic operating scan cycle of the Mitsubishi FX5U PLC?

The PLC executes a repetitive 3-phase scan cycle: (1) Input Scan: Reads physical voltages on terminals X0-X7 into the Input Image Table. (2) Program Execution: Executes ladder rungs sequentially from Step 0 to END, evaluating logic and updating internal registers (M, T, C, D). (3) Output Refresh: Writes the updated output image memory directly to physical output terminals Y0-Y7. Basic instruction execution takes 34ns.

EXAMINER SCORING TIP:
Mention that inputs/outputs do not update mid-scan; they are refreshed synchronously at the start and end of the scan.

Q2: What is the difference between system relays SM400, SM402, and SM412?

SM400 is an 'Always ON' flag (logic TRUE continuously while CPU is in RUN). SM402 is the 'First Scan Pulse' that turns ON for exactly one scan cycle upon STOP->RUN transition, used for initializing recipes (D100-D104). SM412 is a 1.0-second clock pulse generator (0.5s ON, 0.5s OFF), used in Rungs 85 & 92 to blink fault alarm lamps and buzzer Y7 at 1Hz.

EXAMINER SCORING TIP:
Examiners love asking about SM402. Emphasize that without SM402, recipe registers would boot with zero, breaking timer calculations.

Q3: Why did you use Momentary bit switches for Start (M54) and Stop (M55) instead of Alternate?

In industrial safety standards (NFPA 79), start/stop circuits must use momentary 1-shot pushbuttons with seal-in (latching) rungs. If an Alternate (maintained) switch is used for Start and the plant loses power, the machine would dangerously self-start as soon as power is restored. Momentary switches guarantee zero-voltage reset protection.

EXAMINER SCORING TIP:
Safety standard keyword: 'Zero-voltage reset protection / Anti-restart on power recovery'.

Q4: How does the ladder logic detect a conveyor jam fault (M16)?

Rung 14 monitors Conveyor output Y1 and optical sensor X2. When Conveyor Y1 runs, Timer T4 (5.0s) begins counting. If a part reaches X2 before 5 seconds, sensor X2 resets T4. However, if the belt runs continuously for 5.0s without a part reaching X2 (indicating a part is stuck on the belt or belt is slipping), T4 elapses and executes SET M16, stopping conveyor Y1 and sounding buzzer Y7.

EXAMINER SCORING TIP:
Explain that T4 is an 'accumulative delay' protecting against motor stall and chain snapping.

Q5: Explain the two-tier pressure verification architecture between X7, D110, and D102.

The system employs both discrete and analog pressure checking: Physical toggle X7 acts as the header supply switch. Register D110 holds the live line pressure (6.0 bar / 60 counts). Register D102 holds the recipe minimum threshold (4.0 bar / 40 counts). Rung 19 verifies both: [X7] [>= D110 D102] · (M21). If pressure is below recipe requirement or X7 is tripped, M21 stays OFF, halting the spray.

EXAMINER SCORING TIP:
Highlights understanding of software recipe comparison vs physical sensor switching.

Q6: Why must Exhaust Blower Y3 start before Spray Solenoid Y0?

Paint atomization releases high concentrations of volatile organic compounds (VOCs) and solvent vapors which are flammable and toxic. Starting Exhaust Blower Y3 first establishes negative pressure inside the booth. This creates an inward airflow barrier at the openings, preventing solvent fumes from spilling into the factory breathing zone before atomization starts.

EXAMINER SCORING TIP:
Keywords: 'Negative pressure booth containment', 'VOC air exchange', 'OSHA / NFPA 33 compliance'.

Q7: What is the function of Timer T3 (2.0s Pump Settle Delay)?

When paint pump Y2 starts, the fluid line is initially at low pressure. Positive displacement pumps generate hydraulic pulsation during spin-up. If the spray nozzle opens immediately, paint sputters in irregular droplets, ruining the finish. Timer T3 delays opening nozzle Y0 for 2.0s until paint line hydraulic pressure reaches laminar, stable atomization pressure.

EXAMINER SCORING TIP:
Impress the examiner by mentioning 'hydraulic pressure ripple' and 'paint droplet atomization uniformity'.

Q8: How is part tracking and batch clearing synchronized between Counters C0 and C1?

Counter C0 counts infeed parts arriving at sensor X2. When C0 reaches recipe preset D100 (15 parts), Batch Full flag M3 latches and infeed stops. After spraying and purge, finished parts discharge past exit sensor X3, incrementing Outfeed Counter C1. When C1 reaches D100 (15 parts), Cycle Complete pulse M12 triggers: it resets C0, C1, M3, M8, M11, and increments lifetime batch counter C2, readying the booth for the next batch.

EXAMINER SCORING TIP:
Shows that the booth cannot reload until all 15 parts are verified discharged.

Q9: What is the difference between 'TN0' and 'T0' in GT Designer3?

'T0' is a 1-bit boolean device representing the timer contact state (0 = timing, 1 = time elapsed). 'TN0' is a 16-bit word device representing the Timer Current Value (accumulated tick count). A Numerical Display object must point to TN0 to show the live counting numbers; pointing to T0 would cause a data-type mismatch error or show only 0/1.

EXAMINER SCORING TIP:
Directly references the troubleshooting problem solved in earlier commissioning!

Q10: What is the purpose of the Defect Reject Pushbutton (M44) and Counter C4?

If an operator visually spots a warped, dented, or improperly coated part during the outfeed phase, pressing Reject PB M44 triggers latch M46. This energizes pneumatic diverter solenoid Y6 for 1.0s (T7) to kick the defective part into a scrap bin and increments Defect Counter C4. It tracks quality metrics without disrupting the automated batch sequence.

EXAMINER SCORING TIP:
Emphasize Quality Management / OEE scrap tracking.

Q11: Explain the function of Rung 70: [Y0] [Y1] ⋯ (SET M51).

This is the Anti-Collision / Mutual Exclusion Safety Interlock. The spray guns (Y0) and conveyor belt (Y1) must never operate concurrently. If conveyor movement occurs during spraying, parts would receive diagonal streaked coatings and paint would blast onto conveyor chains. Rung 70 instantly latches Abnormal Process Fault M51 if both coils ever fire together, cutting master run and sounding buzzer Y7.

EXAMINER SCORING TIP:
Keyword: 'Mutual exclusion interlock / process integrity monitoring'.

Q12: How does the HMI silence buzzer Y7 without resetting the active fault?

When an alarm trips (M6), buzzer Y7 sounds continuously or pulses with SM412. Pressing SILENCE BUZZER (M52) latches M53. In Rung 84, normally closed contact [/M53] cuts power to buzzer coil Y7. However, the fault latch (e.g. M16, M17, M18) remains active, and pilot lamp Y5 stays lit until the physical defect is corrected and RESET FAULTS (M49) is pressed.

EXAMINER SCORING TIP:
Demonstrates proper industrial alarm acknowledgement philosophy (silence audio, maintain visual alert).

Q13: Why did we use Timer T10 in the ladder program?

Timer T10 is an autonomous, self-resetting 0.8s oscillator (/T10 · OUT T10 K8). It acts as an automatic part indexing simulator: while conveyor Y1 is running, T10 pulses every 0.8 seconds to increment infeed counter C0 and outfeed counter C1, allowing hands-free automated demonstration without having to manually click physical switch X0 or X3 fifteen times per batch.

EXAMINER SCORING TIP:
Shows pragmatic engineering for testability and demonstration.

Q14: What does 'Data Operation: $$ / 10' do in the GT Designer3 properties?

In Mitsubishi FX5U, Timer T0 has a base resolution of 100 milliseconds (0.1s). A 10.0-second timer preset is stored as the integer 100 in D101. When monitoring current value TN0, the raw count displays 0 to 100. Applying the expression '$$ / 10' instructs the HMI display engine to divide the raw register value by 10 before rendering it on screen, cleanly converting 100 raw ticks into '10' seconds for the operator.

EXAMINER SCORING TIP:
Explains integer math conversion without floating-point CPU overhead.

Q15: What is Preventive Maintenance Counter C6 and register D103?

Spray nozzles erode over time due to abrasive pigments, causing orifice wear and pressure drops. Counter C6 tracks completed batches (M12). When C6 reaches Maintenance Setpoint D103 (default 50 batches), Maintenance Due relay M20 latches, pulsing the tower beacon M72 to notify maintenance technicians to inspect nozzles, flush fluid filters, and reset the counter via M48.

EXAMINER SCORING TIP:
Highlights Predictive / Preventive Maintenance (PdM/PM) industry 4.0 capability.

Q16: If the operator opens the chamber door (X5) while spraying, what happens?

Opening the door opens switch X5. In Rung 20, contact [X5] opens, immediately de-energizing Safe to Paint relay M10, which drops Spray Active relay M24 and shuts spray solenoid Y0 in <15 milliseconds. Simultaneously, Rung 47 detects [/X5] while system is in RUN (M0) and latches Chamber Door Open Fault M19, stopping all drives, sounding the horn, and alerting the HMI.

EXAMINER SCORING TIP:
Emphasizes dual response: immediate hardware drop + latched safety diagnostic.

Q17: Why is Alternate action used on Manual Jog Conveyor instead of Momentary?

Momentary mode requires continuous physical contact on the touch panel ('inch' mode). When an operator is commissioning or cleaning the conveyor belt, they need the belt to run hands-free while inspecting tracking and chain alignment. Setting the switch action to Alternate (Toggle) allows one tap to latch conveyor Y1 ON continuously, and a second tap to stop it.

EXAMINER SCORING TIP:
Directly answers the operator usability request implemented in the project!

Q18: What happens if the PLC loses power in the middle of a 15-part batch?

In Mitsubishi FX5U, data registers D100-D110 and counters C0-C6 are battery/latch-backed by default (non-volatile core memory). When power is restored, Counter C0 retains its exact accumulated count (e.g. 8 parts). However, Master Run M0 boots in the OFF state (fail-safe reset). The operator simply taps Start (M54) and the batch resumes from part 9 without losing count or requiring a manual scrap purge.

EXAMINER SCORING TIP:
Keywords: 'Non-volatile retentive memory registers', 'Fail-safe cold boot protection'.

SECTION 6: MASTER SYSTEM MEMORY & DEVICE ALLOCATION TABLE

Quick reference sheet for all inputs, outputs, timers, counters, internal relays, and data registers used across the project.

Device	Type	Assigned Function / Tag	Preset / Value	Screen Location
X0	Input	PB_MAN_FEED: Manual part infeed count pulse	24V Pushbutton	Hardware Kit
X1	Input	PB_STOP: Master system cycle stop	24V Pushbutton	Hardware Kit
X2	Input	LS_PART_IN: Optical sensor 1 (Spray station)	Retro-reflective	Hardware Kit
X3	Input	LS_PART_OUT: Optical sensor 2 (Exit station)	Retro-reflective	Hardware Kit
X4	Input	PB_ESTOP: Emergency stop physical button	NC Safety Contact	Hardware Kit
X5	Input	SW_DOOR: Chamber door closed interlock	SPST Toggle	Hardware Kit
X6	Input	SW_PAINT_LVL: Paint reservoir level normal	SPST Toggle	Hardware Kit
X7	Input	SW_AIR_PRESS: Compressed air pressure normal	SPST Toggle	Hardware Kit
Y0	Output	SOL_SPRAY: Paint atomization spray valve	24V Solenoid	Auto (B-2), Jog (B-3)
Y1	Output	MTR_CONVEYOR: Workpiece infeed/outfeed belt	24V Contactor	Auto (B-2), Jog (B-3)
Y2	Output	PUMP_PAINT: High-pressure paint fluid pump	24V Motor Relay	Auto (B-2), Jog (B-3)

Y3	Output	FAN_EXHAUST: Booth VOC fume extraction blower	24V Fan Relay	Auto (B-2), Jog (B-3)
Y4	Output	LP_RUN: Master system running status pilot lamp	24V Green Pilot	Hardware Kit
Y5	Output	LP_FAULT: General system fault alarm pilot lamp	24V Amber Pilot	Hardware Kit
Y6	Output	LP_ESTOP: Emergency stop active pilot lamp	24V Red Pilot	Hardware Kit
Y7	Output	HORN_BUZZER: Acoustic audio alarm horn	24V Piezo Horn	Hardware Kit
M0	Internal	SYS_RUN: Master system run latch	1=Run, 0=Stop	Auto (B-2)
M7	Internal	FLT_ESTOP: Latched emergency stop condition	Trip Flag	Alarms (B-6)
M16	Internal	FLT_JAM: Conveyor jam timeout fault	5.0s Timeout	Alarms (B-6)
M17	Internal	FLT_PAINT_LOW: Low paint reservoir fault	X6 Low Float	Alarms (B-6)
M18	Internal	FLT_AIR_LOW: Low pressure timeout fault	5.0s Timeout	Alarms (B-6)
M19	Internal	FLT_DOOR_OPEN: Booth door opened fault	X5 Opened	Alarms (B-6)
M51	Internal	FLT_ABNORMAL: Concurrent Y0+Y1 run fault	Mutual Collision	Alarms (B-6)
M49	HMI PB	HMI_RESET: Reset all latched system faults	Momentary	Auto (B-2), Alarms
M52	HMI PB	HMI_SILENCE: Mute alarm audio buzzer Y7	Momentary	Alarms (B-6)
M54	HMI PB	HMI_START: Master cycle auto start	Momentary	Auto (B-2)
M55	HMI PB	HMI_STOP: Master cycle auto stop	Momentary	Auto (B-2)
T0	Timer	TIM_SPRAY: Spray atomization duration	Preset: D101 (10s)	Auto Readout (TN0)
T1	Timer	TIM_START_DLY: Conveyor pre-start delay	Preset: K20 (2.0s)	Internal PLC
T2	Timer	TIM_PURGE: Post-spray VOC fume purge delay	Preset: D104 (5.0s)	Internal PLC
T3	Timer	TIM_PUMP_STBL: Pump pressure settle delay	Preset: K20 (2.0s)	Internal PLC
T4	Timer	TIM_JAM_DLY: Conveyor infeed jam detection	Preset: K50 (5.0s)	Internal PLC
T5	Timer	TIM_PRESS_DLY: Low pressure buildup timeout	Preset: K50 (5.0s)	Internal PLC
C0	Counter	CNT_PARTS_IN: Batch infeed part accumulator	Preset: D100 (15)	Auto Readout (CN0)
C1	Counter	CNT_PARTS_OUT: Outfeed discharged parts verifier	Preset: D100 (15)	Auto Readout (CN1)
C2	Counter	CNT_TOTAL_BATCHES: Lifetime completed batches	Target: K32767	Internal OEE
C3	Counter	CNT_TOTAL_GOOD: Lifetime good parts produced	Target: K32767	Internal OEE
C4	Counter	CNT_DEFECTS: Total defective parts rejected	Target: K32767	Auto Reject M44
C6	Counter	CNT_MAINT: Preventive maintenance interval	Preset: D103 (50)	Maintenance M20
D100	Data Reg	REC_BATCH_SZ: Recipe target batch quantity	Default = 15	Recipe Screen (B-4)
D101	Data Reg	REC_SPRAY_TM: Recipe spray duration (0.1s)	Default = 100 (10s)	Recipe Screen (B-4)
D102	Data Reg	REC_PRESS_MIN: Minimum air pressure setpoint	Default = 40 (4 bar)	Recipe Screen (B-4)
D103	Data Reg	REC_MAINT_INT: Maintenance batch interval	Default = 50	Recipe Screen (B-4)
D104	Data Reg	REC_PURGE_TM: Post-spray purge duration	Default = 50 (5.0s)	Recipe Screen (B-4)
D110	Data Reg	LIVE_PRESSURE: Scaled actual supply air pressure	Default = 60 (6 bar)	Auto Screen Readout